

[BUG] `qml.measure` is not differentiable with postselection with autograd

<https://github.com/PennyLaneAI/pennylane/issues/4813>

ROW 104

[BUG] `qml.measure` is not differentiable with postselection with autograd interface #4813

[New issue](#)🔒 Closed

mudit2812 opened on Nov 9, 2023

Expected behavior

```
Iteration = 0, Energy = 0.00000000 Ha
Iteration = 2, Energy = 0.00000000 Ha
Iteration = 4, Energy = 0.00000000 Ha
Iteration = 6, Energy = 0.00000000 Ha
Iteration = 8, Energy = 0.00000000 Ha
```

Actual behavior

```
TypeError: loop of ufunc does not support argument 0 of type ArrayBox which has no callable sqrt method
```

Additional information

This is because `qml.math.norm` dispatches to `scipy.linalg.norm`, which is not differentiable when the L2 or frobenius norm is being evaluated.

Source code

```
import pennylane as qml
from pennylane import numpy as np
import random

dev = qml.device("default.qubit", wires=2)

layers = 3
n_wires = 2
params = np.random.rand(layers, n_wires, requires_grad=True)

H = qml.Hamiltonian(
    [1 for _ in range(n_wires)],
    [qml.PauliX(i) @ qml.PauliY((i + 1) % n_wires) for i in range(n_wires)]
)

def entangling_block(params):
    for i in range(n_wires):
        qml.RX(0.3, wires=i)
        qml.CRY(params[i] * 2 * i + 1, wires=[i, (i + 1) % n_wires])

@qml.qnode(dev)
def circuit(params):
    for layer_params in params:
        entangling_block(layer_params)
    m_0 = qml.measure(0, postselect= random.randint(0, 1))
    return qml.expval(H)

def cost_fn(params):
    return circuit(params)

opt = qml.GradientDescentOptimizer(stepsize=0.4)
max_iterations = 10

for n in range(max_iterations):
    params, prev_energy = opt.step_and_cost(cost_fn, params)
    energy = cost_fn(params)
```

Assignees

No one assigned

Labels

bug 🐛

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

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Participants



```
# Print the iteration details
if n % 2 == 0:
    print(f"Iteration = {n}, Energy = {energy:.8f} Ha")
```

Tracebacks

No response

System information

```
Name: PennyLane
Version: 0.33.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/PennyLaneAI/pennylane
Author:
Author-email:
License: Apache License 2.0
Location: /Users/mudit.pandey/.pyenv/versions/3.10.12/envs/pennylane/lib/python3.10/site-packages
Editable project location: /Users/mudit.pandey/repos/pennylane
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, requests, rustworkx, s
Required-by: PennyLane-Lightning

Platform info:      macOS-13.6.1-arm64-arm-64bit
Python version:     3.10.12
Numpy version:      1.26.1
Scipy version:      1.11.1
Installed devices:
- lightning.qubit (PennyLane-Lightning-0.32.0)
- default.gaussian (PennyLane-0.33.0.dev0)
- default.mixed (PennyLane-0.33.0.dev0)
- default.qubit (PennyLane-0.33.0.dev0)
- default.qubit.autograd (PennyLane-0.33.0.dev0)
- default.qubit.jax (PennyLane-0.33.0.dev0)
- default.qubit.legacy (PennyLane-0.33.0.dev0)
- default.qubit.tf (PennyLane-0.33.0.dev0)
- default.qubit.torch (PennyLane-0.33.0.dev0)
- default.qutrit (PennyLane-0.33.0.dev0)
- null.qubit (PennyLane-0.33.0.dev0)
```

Existing GitHub issues

☒ I have searched existing GitHub issues to make sure the issue does not already exist.



mudit2812 added **bug** on Nov 9, 2023



mudit2812 on Dec 8, 2023

Contributor

Author

...

Fixed in [#4766](#)



mudit2812 closed this as **completed** on Dec 8, 2023